

Umeshkumar Ghaskata

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University at Buffalo, SUNY(MS in Computer Science)

May 2026

Indian Institute of Technology, Bombay(B.Tech in Aerospace Engineering)

May 2024

PROFESSIONAL EXPERIENCE

DRONES Lab(NASA AIST-21 Project) — ML Engineer [GitHub](#)  [USA](#) [Aug'2025-Jan'2026]

- Designed, trained, evaluated, and deployed **DL models** (PCA+MLP, FPGA-based architectures) for hyperspectral image classification improving land-cover recognition accuracy from 82.1% to 92.2%.
- Reduced **edge inference latency** from **2.7s to 32ms** using optimized FPGA Model for real-time output.
- Built cross-platform benchmarking framework across **RTX 4060, Jetson Orin, and Xavier** to quantify deployment trade-offs and guide edge hardware selection.

Roswell Park Research Corporation — Full Stack Developer [Buffalo, USA](#) [Jan'2026-May'2026]

- Fine-tuned the Gemma 3-1B model on AWS EC2 GPU using QLoRA adapter and **Hugging Face PEFT and deployed to UB server** to build a personalized conversational AI.
- Developed cross-platform mobile applications using **React Native 0.78**, implementing multi-screen navigation, daily check-ins, push notifications, and local storage support for both **iOS and Android**.
- Engineered **CI/CD pipelines** reducing deployment time by **40%**, and optimized **Redux state management** to reduce redundant API calls by **25%**.

Scitara Corporation — Software Developer Intern [Mumbai, India](#) [Dec'2022-Feb'2023]

- Reduced **manual testing time by 30%** implementing virtual server responses to **mock RESTful API responses**, streamlining testing by JSON responses for GET-POST to **waiving credential** requirement

PROJECTS

Multimodal AI Platform with RAG, Streaming [GitHub](#)  [Jan'2026–Mar'2026]

Transformers, LangChain, FAISS, CLIP, Kafka, PySpark, MLflow, FastAPI, Docker, AWS/GCP

- Built a **multimodal AI system** integrating **RAG pipelines**, **Agentic Workflow**, real-time streaming (Kafka), and big data processing (PySpark) for context-aware reasoning across text, documents, and images
- Designed an **end-to-end retrieval and inference pipeline** using CLIP embeddings, FAISS vector search, and LLMs with re-ranking, implementing **MLflow-based** model lifecycle management with automated retraining triggered by data and model drift

Deepface-Similarity-Engine | Image Processing [GitHub](#)  [Jul'2025–Sep'2025]

TensorFlow, Keras, OpenCV, MTCNN, VGGFace (ResNet50), NumPy, Streamlit

- Built a facial recognition system using **MTCNN** for face detection and **VGGFace (ResNet50)** for feature extraction, generating 2048-d embeddings for identity classification
- Developed an end-to-end pipeline with **TensorFlow** and **NumPy**, and deployed an interactive interface using **Streamlit** for real-time image upload and inference

GenAI Full Stack Interview System | AI-Powered Web Application [GitHub](#)  [Jan'2026–Mar'2026]

React.js, Node.js, Express.js, MongoDB Atlas, Google Gemini, JWT, Zod, Puppeteer

- Developed an AI-powered interview preparation platform using **React.js**, **Node.js**, **MongoDB Atlas**, and **Google Gemini**, enabling resume-JD matching, automated skill-gap analysis, ATS-friendly resume generation, and personalized technical interview roadmaps.
- Engineered a production-grade backend with **structured LLM outputs (Zod schema validation)**, JWT authentication, token blacklisting, PDF parsing, and dynamic interview question generation.

TECHNICAL SKILLS SUMMARY

Languages	Python, React-Native, C++, JS, SQL
ML/AI Framework	PyTorch, NLTK, Scipy, TensorFlow, Scikit-learn, Transformers, LangChain, LlamaIndex, RAG, LLM Fine-Tuning, Multi-modal Learning, OpenCV, PySpark, SFT, XGBoost, Hugging face, AXLearn
MLOps/Deployment	MLflow, Docker, CI/CD, FastAPI, Kafka, Model Deployment, Model Drift Detection, AWS, GCP
Database	Redis, FAISS, PostgreSQL, MySQL, Firebase, REST APIs, Distributed Systems, Streaming Pipelines
Certifications	Generative AI Leader(Google)

SCHOLASTIC ACHIVEMENT

- Ranked in the **top 0.7% of 1.3+ million candidates** in national Joint Entrance Examination [2020]